COMMUNITY DISASTER VOLUNTEER HUB

CDVH 20

RAG Knowledge Engineering and Disaster Response Decision Support

Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation
Geographic scope: Technical design for Python, ChromaDB and a CDVH assistant

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Intended audience	Public, volunteers, community coordinators and RAG developers
Use limitation	Educational reference; not a live alert, operational order, professional inspection, medical treatment or legal advice
Research status	Official source pages and source limitations are listed in this document

Safety rule: if a person is in immediate danger or a current evacuation instruction applies, take protective action and contact local emergency services. Do not wait for a chatbot to complete an assessment.

Contents and document use

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How to use this document

Sections are designed to be retrieved independently. Preserve the hazard, geography, time period, conditions and source context when indexing. This PDF is reference material; current warnings, road closures, shelter status and live environmental readings require a current authorized source.

Section map

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Additional components: 24 FAQs, a glossary, references and source/update limitations.

Risk concepts

Susceptibility describes relative tendency for a place to experience a process. Hazard concerns the potentially damaging event and its characteristics over a defined period. Exposure concerns people and assets in harm's way. Vulnerability describes conditions that increase susceptibility to harm. Capacity includes resources and abilities that help reduce consequences. Risk concerns potential adverse consequences considering these dimensions and uncertainty. A susceptibility map is not an imminent-event forecast; a low map class is not a guarantee of safety.

01 Core science and definitions

Document 20 | Section 01 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

Retrieval-Augmented Generation (RAG) retrieves source passages and supplies them to a language model before answer generation. Good RAG requires source quality, document parsing, semantic chunking, metadata, hybrid or reranked retrieval where useful, prompt controls and evaluation. Retrieval is not verification by itself: a relevant but outdated or misapplied passage can still produce an unsafe answer.

In a RAG system, the equivalent distinction is between a source document, a retrieved passage, a model-generated synthesis and a verified operational fact. These are not interchangeable. Preserve source identity and date with every chunk; retrieval indicates relevance, not truth, freshness or authority.

For a retrieved passage about retrieval grounding, knowledge maintenance, citations and safety evaluation, the minimum useful context is: what process is being discussed, where the statement applies, whether it is general or current, who issued the information, and what practical decision the reader is trying to make. A passage missing one of these may still explain a concept, but it should not be used alone for an operational decision.

For engineering purposes, a retrievable chunk should be self-contained but not stripped of source lineage. Store a stable section ID, hazard/domain, geographic scope, applicable conditions, source URL, access date, version and safety-critical flag beside the passage text.

Decision aid

Term	Meaning for decision support	Do not infer
Hazard / process	What could cause harm and how it may operate	That harm is occurring at a named house now
Exposure	People, buildings, services or livelihoods in the affected area	That all exposed people have equal vulnerability
Vulnerability / capacity	Conditions that can increase harm or improve coping	That a map or average describes every person
Risk statement	Potential consequences over a stated period and context	An exact probability without suitable evidence

Practical checks

- Write the geography and time window near the claim.
- Attribute official instructions to the issuing authority and preserve issue time.
- Keep uncertain inference separate from observation and published guidance.

Supporting source identifiers: [S01], [S04], [S05], [S06]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

02 Causes, contributing factors, and mechanisms

Document 20 | Section 02 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

Failures include missing pages, lost table headings, oversized chunks, duplicate content, inconsistent metadata, embedding mismatch, stale warnings, conflicting sources, poor question decomposition, low recall, prompt injection embedded in source text and unsupported synthesis. Vector similarity is a relevance signal, not an authoritative confidence probability.

A useful explanation separates background susceptibility, a trigger, a pathway of impact and exposed assets. The word 'cause' should be used with care: several conditions may contribute, and the data available to a chatbot may not establish which factor controlled an individual event. The relevant scientific mechanism for retrieval grounding, knowledge maintenance, citations and safety evaluation must be matched to the actual site or system rather than assumed from a neighbouring district.

When explaining a possible event, show how evidence links each stage. For example, a regional weather event may create a broad loading or stress condition, but local terrain, drainage, construction, infrastructure condition or exposure can change the result. In other topics, the same logic applies to river-network routing, coastal water levels, building response, fire spread or gaps in a technical data pipeline. A plausible pathway is not proof that the pathway will occur.

Common failure paths include extracting a PDF without headings, embedding table fragments, losing dates, mixing old and current versions, and allowing the model to infer unsupported causality from adjacent chunks. Evaluation must test these failure modes deliberately.

Decision aid

Stage	Question to ask	Evidence limitation
Precondition	What makes this place or system susceptible?	May be incompletely mapped or not visible
Trigger	What recent forcing or failure could initiate change?	A trigger rarely determines the exact outcome alone
Pathway	How could the process reach people/assets?	Terrain, network or building connection matters
Consequence	Who or what could be affected, and how?	Exposure and protective capacity vary locally

Practical checks

- Use 'may contribute' when the evidence is conditional.
- Do not assert a single cause from timing alone.
- Route technical mechanism questions to qualified authorities when site-specific judgment is necessary.

Supporting source identifiers: [S01], [S04], [S05], [S06]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

03 Indicators, warning signs, and limitations

Document 20 | Section 03 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

Track whether retrieved passages include the correct hazard, geography, conditions, date, source and action; whether references are precise; whether the answer adds unsupported facts; and whether emergency instructions appear first. Monitor zero-result queries, disagreement among sources, stale alert retrieval and repeated low-quality answers.

Warning signs are observations that can justify protective action or reporting; they are not a complete diagnostic test. The evidence may be early, ambiguous or absent. A checklist cannot promise that an event will be detected before it becomes dangerous. Users should observe from a safe location and report what they saw, where and when, without approaching the hazard to obtain a clearer picture.

The safest public workflow is to separate direct observation from a forwarded message, a model interpretation and an official bulletin. Record the time, approximate place, direction of change and whether the observation is continuing. If the sign suggests immediate danger, first move away by a safe route and alert appropriate emergency services; the chatbot should not ask for photos or multiple confirmations before recommending that action.

For RAG QA, label each test question with the evidence it requires, the safe action, sources that should be retrieved, and claims that must not be generated. Cases involving missing alerts, stale pages and conflicting sources are particularly important regression tests.

Decision aid

Evidence type	Useful interpretation	Safe next action
Direct, changing physical sign	May indicate a developing local danger	Move away if threatened; report from safety
Official alert or order	Authoritative for its stated area/time and instruction	Follow it; verify updates as appropriate
Third-party message / screenshot	Unverified until source and timestamp are checked	Do not forward as fact; verify through official channel
No visible sign / no alert	Absence of evidence, not proof of safety	Use other relevant evidence and current guidance

Practical checks

- Never enter, cross or approach a hazard to verify a sign.
- Do not require every listed sign to be present.
- Explain when signs are nonspecific and why qualified assessment may be needed.

Supporting source identifiers: [S01], [S04], [S05], [S06]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

04 Location-specific risk assessment

Document 20 | Section 04 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

Use stable document_id and section_id, source agency/title/url/publication/access dates, document version, geography, hazard, content type, safety-critical flag and supersession status as metadata. Evaluate recall@k, precision or relevance, citation correctness, answer completeness and safety. Keep live alert data separate from static guidance and check their timestamps independently.

A preliminary risk screen should answer four distinct questions: what is the hazard or process; what present evidence describes it; which people or assets may be exposed; and what conditions make the consequences more or less severe. Static maps and regional alerts are inputs, not individual guarantees. State the observation date, map scale or forecast period, geographic coverage and uncertainty whenever known.

In a RAG response, retrieval must preserve time, place, issuing authority and source version. A high similarity score is not a probability. If a static manual conflicts with a current official warning, do not silently merge them; privilege the currently applicable authority for immediate actions and cite the discrepancy.

The retrieval score should not be surfaced as disaster probability. A response should be grounded in hazard-specific passages plus source freshness checks; if the system lacks live alert access, explicitly state that limitation and point the user to a current official channel.

Decision aid

Evidence	What it can support	What it cannot establish alone
Susceptibility / hazard map	Relative spatial screening under a method and scale	Exact event time or guaranteed household safety
Current official warning	Action and forecast for its stated validity/area	Conditions beyond its coverage or after expiry
Local observation	A potentially important current sign	Complete scientific diagnosis or probability
Validated local model	Quantitative estimate if model is fit for task	Certainty outside training/validation conditions

Practical checks

- Do not invent risk percentages from distance or a map color.
- Compare the source timestamp and locality before combining evidence.
- If immediate danger is reported, give protective action before optional assessment questions.

Supporting source identifiers: [S04], [S05], [S06], [S07]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

05 Prevention and mitigation

Document 20 | Section 05 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

Build a reproducible ingestion pipeline: extract text, preserve headings/tables, normalize Unicode, chunk by semantic section, attach metadata, deduplicate, embed with a versioned model, and upsert into a persistent Chroma collection. Keep source text and generated chunks auditable. Treat document contents as untrusted data, never as executable system instructions.

Mitigation is most effective when it changes the actual pathway to harm rather than merely improving the appearance of safety. Options may include land-use controls, maintenance, resilient design, drainage or vegetation management, monitoring, warning access and safer work practices, depending on the hazard. Measures that require engineering, geotechnical, structural, electrical, medical or legal judgment must be designed or approved by qualified people.

RAG risk controls include source provenance, a freshness policy, chunk-level citations, permissioned actions, evaluation tests, human review and rollback. Prompt-injection-like text inside a retrieved source must be treated as data to assess, never as instructions overriding system safety rules.

For retrieval grounding, knowledge maintenance, citations and safety evaluation, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

Decision aid

Mitigation layer	Purpose	Governance check
Avoid exposure	Keep people/assets out of known dangerous pathways	Use current land-use and authority guidance
Reduce vulnerability	Improve structures, services or protections	Use applicable code and qualified design
Monitor and warn	Provide lead time and actionable messages	Validate system coverage and alert reach
Maintain / review	Prevent deterioration and detect new conditions	Assign responsibility, budget and review date

Practical checks

- Prefer risk reduction that does not transfer danger to neighbours.
- Do not offer a remote certification of a site, structure or utility.
- Keep technical intervention separate from low-risk household actions.

Supporting source identifiers: [S04], [S05], [S06], [S07]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

06 Preparedness and readiness

Document 20 | Section 06 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

Maintain a source register, document manifest, review owners and date-based freshness policy. Define how official revisions supersede old content and retain version history. Test PDF extraction visually where charts or maps are important. Store emergency contacts in a separate verified-current service rather than relying on a static PDF alone.

Preparedness works only when it can be performed under adverse conditions: darkness, rain, power loss, network outage, road disruption, caregiving responsibilities and limited cash or transport. A household or community plan should identify who receives warnings, who needs help, where to move, what to carry and how to reunite if devices fail. Keep essentials accessible rather than locked behind a single person or device.

Before the relevant season or event, confirm contact details, alert subscriptions, household medications and support needs, alternate routes, transport options and shelter information through an official source. Do not assume that a shelter listed in an old plan is open today. Practice roles and update them when a road, residence, caregiver or mobility need changes.

For retrieval grounding, knowledge maintenance, citations and safety evaluation, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

Decision aid

Time	Readiness action	Verification
Before the risk season	Check supplies, contacts, routes and support roles	Use current authority/shelter information
At warning stage	Charge devices, monitor updates and prepare to move	Confirm geographic coverage and issue time
When instructed to act	Take the stated protective action promptly	Do not wait for optional chatbot analysis
After an exercise/event	Record what failed and revise the plan	Set an owner and due date for each fix

Practical checks

- Make the plan accessible to children, older adults and people with disabilities.
- Include a no-network communication method.
- Keep sensitive medical or identity details private; share only what is needed for assistance.

Supporting source identifiers: [S04], [S05], [S06], [S07]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

07 Immediate response procedures

Document 20 | Section 07 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

At answer time detect emergencies, retrieve hazard-specific and safety policy passages, check if live data are needed, query only the available authorized integrations, then generate a concise answer with cited source dates and limitations. If required evidence is absent, say so, offer safe default advice and escalate to qualified authorities.

Response advice must be ordered by urgency. Protect life and avoid turning a bystander into another casualty; use the appropriate emergency channel; follow current official instructions; communicate location and observations from safety; and leave technical tasks to qualified responders. The chatbot should not claim it dispatched help, checked live conditions or sent an alert unless an authorized integration confirms that action.

When a user reports a rapidly worsening condition, keep the initial response short and direct. Give the first safe action before explaining scientific background or asking for a detailed location. A broad location can help identify the relevant authority, but exact coordinates should not be required to recommend avoiding an immediate hazard. Once the person is safe, gather only additional details that affect the next decision.

For retrieval grounding, knowledge maintenance, citations and safety evaluation, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

Decision aid

Priority	Action	Boundary
1. Protect	Move away from immediate danger by a safe route	Do not cross the hazard or retrieve belongings
2. Alert	Contact current local emergency/response channels	Do not imply the chatbot called on the user's behalf
3. Inform	Give approximate location, time and observed condition	Report facts separately from assumptions
4. Reassess	Follow responders and current updates	Do not return or begin technical rescue without clearance

Practical checks

- Do not delay action to complete a risk score.
- Do not enter unstable slopes, floodwater, smoke, rubble or electrically unsafe areas.
- If the route is compromised, report the obstacle from safety and follow authorized instructions.

Supporting source identifiers: [S04], [S05], [S06], [S07]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

08 Evacuation and protective actions

Document 20 | Section 08 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

Do not present route suggestions as safe unless an authoritative current routing/closure source supports them. For emergency questions, the generation layer must prioritize evacuation orders and immediate warning signs, not maximize detail. Tool failure, empty retrieval or stale live feeds should produce explicit limitations, never fabricated successful checks.

Evacuation depends on timing, route condition, destination and the person's ability to move. A familiar route can become dangerous because of the same event that triggered the evacuation. Use current official route instructions and never infer that a route is safe because it was usable yesterday. If movement is ordered, arrange assistance early and do not wait for a perfect checklist when a threat is immediate.

A decision to shelter in place is hazard-specific and should not be invented by the chatbot. It may be safer for some threats and unsafe for others. The answer must distinguish an official instruction to evacuate, an instruction to remain indoors, a blocked route and a person who sees immediate danger. If an official destination is inaccessible, contact the responsible authority rather than improvising through a dangerous area.

For retrieval grounding, knowledge maintenance, citations and safety evaluation, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

Decision aid

Situation	Safer response principle	Do not
Explicit evacuation order	Follow it using the stated route and timing	Delay to pack nonessential items
Immediate visible danger	Move away if a safe path exists and alert responders	Wait for a model or all warning signs
Route appears blocked	Stop before entering the hazard and report it	Cross floodwater or unstable ground
All-clear not received	Remain cautious and follow official updates	Return because conditions look calm

Practical checks

- Plan for essential medicines, mobility aids, children, older adults and pets where relevant.
- Keep family meeting and reunification arrangements simple.
- Do not promise that a shelter has space without current confirmation.

Supporting source identifiers: [S04], [S05], [S06], [S07]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

09 Volunteer and responder coordination

Document 20 | Section 09 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

Operational controls should include access restrictions, backups, test and production collections, source-change monitoring, prompt/version logs, red-team testing, clear ownership and rollback. Keep private user questions and precise locations out of public embeddings unless justified and consented. Separate document retrieval from actions that send messages, create tasks or notify responders.

Community volunteers are most effective when they fill defined support roles and remain within training, equipment and supervision. Proximity to an incident, local familiarity or good intentions do not confer technical rescue capability. Each task needs an assigning supervisor, boundaries, communication method, buddy or accountability procedure, stop-work trigger and safe withdrawal plan.

Examples of safer support may include relaying verified messages, helping an authorized distribution point, translating warnings, assisting with registration under privacy rules, checking on neighbours from safe locations and reporting obstructions. Unsafe self-assigned activities include entering unstable structures, approaching an active landslide, wading into floodwater, going near live wires or attempting rescue from fire, smoke or chemical release. Escalate specialist work to trained responders.

For retrieval grounding, knowledge maintenance, citations and safety evaluation, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

Decision aid

Task type	Volunteer boundary	Stop / withdraw when
Information relay	Repeat authorized messages with timestamp and area	Message is unverified or outside assigned role
Logistics / welfare	Work at an authorized safe location	Crowding, contamination or scene hazard increases
Observation / reporting	Observe from safety; report facts and time	Closer approach is needed to get more evidence
Technical rescue	Only trained, equipped, authorized teams	Any task exceeds training, command or PPE

Practical checks

- Check in, remain accountable and report back before leaving an assignment.
- Use clear identification without impersonating an official.
- Keep emergency access routes clear and protect affected people's privacy.

Supporting source identifiers: [S01], [S04], [S05]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

10 Vulnerable populations and inclusive planning

Document 20 | Section 10 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

Evaluate language variants, misspellings, low-literacy questions, assistive technologies and accessibility of retrieved PDFs. Avoid overconfidence for underrepresented locations or communities. Ask only for location resolution necessary to answer and do not store it by default.

Inclusive risk communication must fit the person, not just the average user. An alert is ineffective if it cannot be heard, seen, understood, reached in time or acted upon. Plan with people who have mobility, sensory, cognitive, language, health, caregiving or transport needs. Avoid assumptions based on age or diagnosis; ask what practical help is needed and preserve the person's agency and privacy.

For Technical design for Python, ChromaDB and a CDVH assistant, plan several ways to communicate and move: audible and visual messages, plain language, local-language translations, trusted community networks, accessible transport, caregivers, mobility aids, backup charging and alternatives for people without smartphones. The plan should consider medication continuity and service animals where applicable, without asking users to disclose unnecessary private health information in a public chatbot.

For retrieval grounding, knowledge maintenance, citations and safety evaluation, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

Decision aid

Need / barrier	Planning measure	Check
Mobility or transport	Accessible transport, assistance role and a route without avoidable barriers	Can the person use the route in time?
Hearing / vision	Text, audio, visual and person-to-person warning options	Is the message perceivable and understandable?
Language / literacy	Plain language, local translation, tested instructions	Can the person explain the action back?
Health / caregiving	Medicine continuity and caregiver coordination	Can essential support travel with the person?

Practical checks

- Offer choices and confirm consent when it is possible.
- Do not publish home coordinates, diagnosis or beneficiary data openly.
- Treat inaccessible warning reach as a system failure, not user failure.

Supporting source identifiers: [S01], [S04], [S05]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

11 Recovery and lessons learned

Document 20 | Section 11 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

When an unsafe answer occurs, preserve a minimal privacy-safe incident record, identify source/retrieval/prompt/model failure, repair the corpus or policy, add a regression question and rerun the suite. Version the fix and make clear what has or has not been verified. Human domain experts should review high-risk safety updates.

Recovery should move from immediate relief to safe restoration, rehabilitation and longer-term risk reduction. A cleaned room, reopened road or absent smoke does not prove that a structure, slope, water source or utility is safe. Re-entry and reconstruction decisions may require formal inspection, testing, repair standards or permission from the responsible authority. Record which decision is confirmed and which remains provisional.

A fair recovery system documents needs and assistance transparently while protecting privacy. Track who has received support and who has not been reached, distinguish estimated losses from verified assessments, and provide accessible complaint routes. Recovery planning should consider housing, water, sanitation, health, education, livelihoods, environment and psychosocial support together instead of treating each as a separate checklist.

For retrieval grounding, knowledge maintenance, citations and safety evaluation, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

Decision aid

Stage	Main concern	Evidence / safeguard
Immediate relief	Safe shelter, water, health, information and protection	Current needs and official coordination
Damage assessment	Document loss without entering unsafe places	Source, date, assessor and uncertainty
Re-entry / repair	Confirm hazards and systems are safe enough	Authorized inspection or testing when required
Build back better	Reduce repeated exposure and vulnerability	Qualified design, community input and maintenance

Practical checks

- Do not guarantee compensation eligibility or payment timing without current rules.
- Keep survivor records secure and collect only necessary data.
- Review whether rebuilding reduces risk rather than recreating it.

Supporting source identifiers: [S01], [S04], [S05]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

12 Complex scenarios and chatbot guardrails

Document 20 | Section 12 | Hazard/domain: retrieval grounding, knowledge maintenance, citations and safety evaluation | Scope: Technical design for Python, ChromaDB and a CDVH assistant

A retriever returns a 2016 disaster plan and a current warning page with different contact information. The system should prefer the currently authoritative and timestamped source for the immediate contact task, flag that older plans may be superseded, cite sources near claims and avoid merging contradictory details silently.

The scenario should be handled with an evidence-first response. Separate what the user directly observed, what the official source says, what the model can infer and what remains unknown. Offer the safest supported next step first; then explain which additional evidence would materially change the assessment. Never convert a similarity to a past case, a nearby incident, a map color, a sensor threshold or a vector score into a guaranteed outcome.

Expected answer: prefer the current applicable authoritative notice for the immediate task, flag any older source as potentially superseded, preserve both source dates, and avoid silently combining incompatible contact details or procedures.

For retrieval grounding, knowledge maintenance, citations and safety evaluation, apply the section's principle to the actual mechanism, location and time horizon. Record what is observed, what is sourced, what is inferred and what is unknown. Do not extend a general rule beyond its stated scope; use local official instructions for decisions that depend on present conditions.

Decision aid

Response element	What a safe answer should include	What it must avoid
Immediate action	Protective step if danger is plausible now	Waiting for analysis or more signs
Known evidence	User observation and current applicable official source	Treating a screenshot or old PDF as live
Uncertainty	Specific unknowns and their relevance	Invented probability or false reassurance
Follow-up	One detail that materially changes guidance	Unnecessary precise location or private data

Practical checks

- State the limit of the available information plainly.
- Use a current official source when time-sensitive information is required and available.
- If no live check occurred, say it was not checked.

Supporting source identifiers: [S01], [S04], [S05]. Full titles, publishers, dates, URLs and limitations appear in References and source notes.

Frequently asked questions

General guidance only; follow current instructions from responsible authorities.

Q1. Does high vector similarity prove a passage is correct?

No. Similarity estimates textual relevance, not truth, freshness or applicability. Validate source, date, geography and context. Source trail: [S01], [S04], [S05], [S06]

Q2. How should chunks preserve citations?

Store source and document metadata with every chunk, keep stable IDs and include source identifiers in the answer context. Source trail: [S01], [S04], [S05], [S06]

Q3. Should static PDFs answer live-alert questions by themselves?

No. Static documents are reference material. Use an authorized current source when available and state when live data were not checked. Source trail: [S01], [S04], [S05], [S06]

Q4. What are the main factors for assessing retrieval grounding, knowledge maintenance, citations and safety evaluation risk?

Use stable document_id and section_id, source agency/title/url/publication/access dates, document version, geography, hazard, content type, safety-critical flag and supersession status as metadata. Evaluate recall@k, precision or relevance, citation correctness, answer completeness and safety. Keep live alert data separate from static guidance and check their timestamps independently. Source trail: [S01], [S04], [S05], [S06]

Q5. Which warning signs matter for retrieval grounding, knowledge maintenance, citations and safety evaluation?

Track whether retrieved passages include the correct hazard, geography, conditions, date, source and action; whether references are precise; whether the answer adds unsupported facts; and whether emergency instructions appear first. Monitor zero-result queries, disagreement among sources, stale alert retrieval and repeated low-quality answers. Source trail: [S01], [S04], [S05], [S06]

Q6. What is the main immediate response for retrieval grounding, knowledge maintenance, citations and safety evaluation?

At answer time detect emergencies, retrieve hazard-specific and safety policy passages, check if live data are needed, query only the available authorized integrations, then generate a concise answer with cited source dates and limitations. If required evidence is absent, say so, offer safe default advice and escalate to qualified authorities. Source trail: [S01], [S04], [S05], [S06]

Frequently asked questions (continued)

Document 20 | Questions 7-12

Q7. How should a household prepare for retrieval grounding, knowledge maintenance, citations and safety evaluation?

Maintain a source register, document manifest, review owners and date-based freshness policy. Define how official revisions supersede old content and retain version history. Test PDF extraction visually where charts or maps are important. Store emergency contacts in a separate verified-current service rather than relying on a static PDF alone. Source trail: [S04], [S05], [S06], [S07]

Q8. What can volunteers safely do during retrieval grounding, knowledge maintenance, citations and safety evaluation?

Operational controls should include access restrictions, backups, test and production collections, source-change monitoring, prompt/version logs, red-team testing, clear ownership and rollback. Keep private user questions and precise locations out of public embeddings unless justified and consented. Separate document retrieval from actions that send messages, create tasks or notify responders. Source trail: [S04], [S05], [S06], [S07]

Q9. How should people needing additional assistance be included?

Evaluate language variants, misspellings, low-literacy questions, assistive technologies and accessibility of retrieved PDFs. Avoid overconfidence for underrepresented locations or communities. Ask only for location resolution necessary to answer and do not store it by default. Source trail: [S04], [S05], [S06], [S07]

Q10. What should happen after the event?

When an unsafe answer occurs, preserve a minimal privacy-safe incident record, identify source/retrieval/prompt/model failure, repair the corpus or policy, add a regression question and rerun the suite. Version the fix and make clear what has or has not been verified. Human domain experts should review high-risk safety updates. Source trail: [S04], [S05], [S06], [S07]

Q11. Does distance alone prove safety or danger?

No. Distance by itself does not establish risk. Evaluate the hazard-specific spatial connection, local conditions, time window, official warning and relevant observations. Source trail: [S04], [S05], [S06], [S07]

Q12. Can an exact probability be given from these documents?

Only if a suitable validated, current and location-relevant model or authoritative forecast explicitly supports it. Otherwise the system must state uncertainty and avoid invented percentages. Source trail: [S04], [S05], [S06], [S07]

Frequently asked questions (continued)

Document 20 | Questions 13-18

Q13. Does the absence of an official warning guarantee safety?

No. Monitoring and communication can be incomplete or delayed. Credible direct signs and immediate danger require protective action and contact with the responsible authority. Source trail: [S04], [S05], [S06], [S07]

Q14. What information should a follow-up question request?

Only details that change advice, such as broad locality, time window, relevant slope/river/coastal connection, current signs and whether danger is happening now. Avoid unnecessary personal data. Source trail: [S04], [S05], [S06], [S07]

Q15. What if maps and observed conditions disagree?

Do not dismiss immediate danger because a static map is reassuring. Report from safety, follow current official instructions and note map date, scale and uncertainty. Source trail: [S04], [S05], [S06], [S07]

Q16. What if current alerts cannot be accessed?

State explicitly that live alerts were not checked, provide conservative general safety guidance and direct the user to the relevant official source or emergency service. Source trail: [S04], [S05], [S06], [S07]

Q17. Can a photograph prove that a site or structure is safe?

No. Images may omit key features and cannot substitute for site inspection, qualified assessment or live situational information. Source trail: [S01], [S04], [S05]

Q18. Where can the public find authoritative information?

Start with the relevant official Indian/state/district authority and technical agency listed in the source register; verify the publication date, geographic coverage and current status. Source trail: [S01], [S04], [S05]

Frequently asked questions (continued)

Document 20 | Questions 19-24

Q19. Can old case studies be used as a forecast?

No. Case studies describe past conditions and lessons, not the exact timing or outcome of a future event. Source trail: [S01], [S04], [S05]

Q20. What should users avoid doing?

Do not enter or approach dangerous areas, cross floodwater, inspect unstable slopes, enter damaged buildings or exceed volunteer training. Follow official instructions. Source trail: [S01], [S04], [S05]

Q21. How should conflicting official messages be handled?

Compare issuing agency, timestamp, geography and scope. Follow the current directly applicable emergency instruction while seeking clarification without delaying life safety. Source trail: [S01], [S04], [S05]

Q22. Can the chatbot confirm that emergency services were called?

Only when an authorized connected service actually confirms the action. Otherwise it should direct the user to call the service directly. Source trail: [S01], [S04], [S05]

Q23. Are the PDFs a live monitoring system?

No. They are static reference documents. Current forecasts, alerts, road closures and shelter availability require a verified live integration or direct official check. Source trail: [S01], [S04], [S05]

Q24. What is a common unsafe assumption about retrieval grounding, knowledge maintenance, citations and safety evaluation?

A common unsafe assumption is treating incomplete information as certainty. A retriever returns a 2016 disaster plan and a current warning page with different contact information. The system should prefer the currently authoritative and timestamped source for the immediate contact task, flag that older plans may be superseded, cite sources near claims and avoid merging contradictory details silently. Follow current official guidance and disclose uncertainty. Source trail: [S01], [S04], [S05]

Glossary and interpretation rules

Hazard: A potentially damaging process, phenomenon or human activity.

Exposure: People, infrastructure, housing, production capacity and other tangible human assets in hazard-prone areas.

Vulnerability: Physical, social, economic or environmental conditions that increase susceptibility to harm.

Susceptibility: Relative tendency of an area or system to experience a process; not a forecast of immediate occurrence.

Capacity: Resources, systems, relationships and abilities available to manage a hazard and reduce harm.

Forecast: A statement about possible future conditions over a specified place and time, based on data and a method.

Official warning: A communication from a responsible authority describing a hazard and recommended action for stated coverage and validity.

Evacuation: Protective movement away from a threatened area by an appropriate route.

Residual risk: Risk remaining after prevention or mitigation; never automatically zero.

Uncertainty: Limits in data, model assumptions, observations, timing, geography or inference.

RAG: Retrieval-Augmented Generation: retrieval supplies source passages that help ground a model response.

Source provenance: Information identifying the origin, issuer, date, version and context of a claim.

Site-specific assessment: Evaluation using local conditions, current evidence and the competence needed for the decision.

References and source notes

Document CDVH-20-KB | Version 1.1.0 | Access date 2026-10-16. Official sources may change; re-check currentness before operational use. The source list is not a claim that every linked attachment was completely extracted and reviewed.

[S01] Kerala State Disaster Management Authority guidelines index - KSDMA. Publication/revision date: Index includes 2026 Orange Book of Disaster Management, Edition 8 (individual attachment date must be checked). URL: https://sdma.kerala.gov.in/guidelines/. Accessed 2026-10-16.

Use and limitation: Current index lists the 2026 Orange Book and other guidance; individual linked documents require independent version review.

[S04] Standard Operating Procedure for Flood Forecasting - Central Water Commission. Publication/revision date: April 2020. URL: https://cwc.gov.in/sites/default/files/sopapril2020final.docx.pdf. Accessed 2026-10-16.

Use and limitation: April 2020 SOP reviewed through its official source page; station lists and contacts may be out of date.

[S05] Overview: Rainfall-induced landslides - U.S. Geological Survey. Publication/revision date: 2018-07-02. URL: https://www.usgs.gov/programs/landslide-hazards/science/overview-rainfall-induced-landslides. Accessed 2026-10-16.

Use and limitation: Explains intense/prolonged rainfall, shallow failures and debris-flow connections; local thresholds are site-dependent.

[S06] What are signs of landslide development? - U.S. Geological Survey. Publication/revision date: Not confirmed during this build. URL: https://www.usgs.gov/programs/landslide-hazards/what-are-signs-landslide-development-what-do-i-do-if-a-landslide-occurs. Accessed 2026-10-16.

Use and limitation: Warning signs are not exhaustive and sudden failures may occur with little visible warning.

[S07] Disaster risk terminology - UN Office for Disaster Risk Reduction. Publication/revision date: Not confirmed during this build. URL: https://www.undrr.org/terminology/disaster-risk. Accessed 2026-10-16.

Use and limitation: Terminology supports hazard, exposure, vulnerability and capacity framing.

[S11] International First Aid, Resuscitation, and Education Guidelines - International Federation of Red Cross and Red Crescent Societies. Publication/revision date: 2025 edition listed on source page; page updated 2026-03-18. URL: https://www.ifrc.org/document/international-first-aid-resuscitation-and-education-guidelines. Accessed 2026-10-16.

Use and limitation: Official IFRC page identifies 2025 guidelines; specific medical protocols must follow local training and updated clinical standards.

[S12] Floodwater after a disaster or emergency: safety guidance - U.S. Centers for Disease Control and Prevention. Publication/revision date: Reviewed August 2026. URL: https://www.cdc.gov/floods/safety/floodwater-after-a-disaster-or-emergency-safety.html. Accessed 2026-10-16.

Use and limitation: Reviewed August 2026 according to the source page; U.S.-specific health guidance adapted only as general principles.

[S18] India Meteorological Department main portal - India Meteorological Department. Publication/revision date: Not confirmed during this build. URL: https://mausam.imd.gov.in/. Accessed 2026-10-16.

Use and limitation: Use for current official forecasts and warnings; this knowledge library does not itself provide live alert data.

[S19] Indian Tsunami Early Warning System - Indian National Centre for Ocean Information Services. Publication/revision date: Not confirmed during this build. URL: https://tsunami.incois.gov.in/TEWS/. Accessed 2026-10-16.

Use and limitation: Official operational tsunami warning system portal; check current notices and location-specific instructions.

[S20] Emergency Response Support System 112 - Government of India. Publication/revision date: Not confirmed during this build. URL: https://112.gov.in/. Accessed 2026-10-16.

Use and limitation: Official nationwide emergency response portal; test/verify local access practices before publishing local workflows.

Citation and update policy

Source identifiers are printed near relevant content and map to this list. For operationally consequential claims, verify the original source, its current status, publication/revision date, geography and relevant section. Do not invent page numbers or infer that a linked attachment was reviewed where the build record says it was unavailable.

Research limitation

This is an educational draft reference corpus, not a formally certified emergency manual. Some official web pages were inspected, but not every listed attachment was downloaded and fully extracted. The source register in this PDF records known limitations. Expert review remains necessary for medical, engineering, geotechnical, GIS, legal and operational content. Static PDFs do not provide live alerts or site-specific predictions.